

Automotive Commodity Chains and the Reproduction of Labor:

A Global Perspective

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Hypothesis

Prevailing wisdom holds that Fordism was the epitome of 20th century American capitalism. Its logic of rationalized production combined with working class consumerism was perfected here during the world wars and then was exported elsewhere, first to Europe after the wars, and then in modified form to the periphery in the 60s and 70s. Fordism's defining characteristics were the rationalization of labor and an increased interest in the social lives of workers and their patterns of consumption. However, attempts to study the origin of this relationship between production and consumption often limit the investigation to American automotive factories and thereby obscure the extent to which early Fordist policies were dependent on extractive practices in the Global South. This project takes a closer look at the relationship between the Fordist policies of the early automotive industry and the conditions of production in the Global South that made them possible. I will look at the connections between social-reproductive relations in Michigan auto factories, and those in the wild rubber fields in the Amazon, and the rubber plantations in Colonial Malaya between 1914 and 1935. I contend that the social relations of Fordism, and especially the organization of workers into home-owning nuclear families, were only made possible through the destruction and reconfiguration of household dynamics in the Global South. Whereas the development of Fordism resulted in an upper echelon of working class

nuclear families in America, in the rubber fields of the Amazon it restructured or dissolved existing familial bonds through the theft of indigenous children and the creation of patterns of transhumance in which workers were forced to travel hundreds of miles from their families during the harvesting season. In the rubber plantations of Colonial Malaya, migrant workers from south India left their families behind in Madras for several years at a time in order to work as rubber harvesters. In this project, I plan to investigate how the organization of housing and living conditions in the periphery, including the de/valuation of reproductive labor and the racial stratification of workers, were manipulated by capital and the state to produce specific types of laborer, and how this in turn made Fordist social and productive relations in the US possible. By examining labor and social archives in Michigan, South America, and Malaya, I hope to establish a definite causal link between the social reproductive policies of the core and periphery along the automotive commodity chain.

Significance

This project will make two primary contributions. The first is to challenge the prevailing literature on Fordism which ignores its foundation of extraction in the Global South. Prominent studies of Fordism (such as Aglietta (1979) and Harvey (1989)) treat it as a phenomenon totally contained within the US and Europe. In this dominant narrative, Henry Ford developed a system of production based on the rationalization and decomposition of labor combined with the development of a commodity consuming workforce organized in nuclear families. This package of a workforce that both produces and consumes was modified slightly when exported to other countries, but it remained

fundamentally a system of family based workers who were paid well enough to constitute a robust domestic market.

My project challenges the view that Fordism originated and developed solely within the US. Rather, I insist that Fordist production stood on the shoulders of the Global South, and was born in the relationship between US factory production and the extraction of raw materials elsewhere. Although Fordist production practices quickly spread to other industries, I take the early automotive industry as exemplary of this global process. Automobiles are not solely the product of the factories they roll out of. Rather, like all commodities, they are the result of a chain of linked production processes that span the globe. This commodity chain includes the final assembly of the car in US cities, but also the harvesting and processing of rubber in the Amazon and Malaya, and the mining of iron and copper in Michigan's upper peninsula; it includes the transportation of materials from extraction sites to processing and manufacturing sites; and finally it includes the labor itself and the conditions that support it: food, shelter, education, and so on. The conditions of production at one point in the commodity chain cannot be separated from those at another point: they form a single process. By examining the conditions of rubber harvesting and cultivation in the Amazon and Colonial Malaya, I hope to demonstrate that the development of a consuming community of workers in automotive factories in the US was made possible by a system of community dissolution and harsh exploitation in other parts of the world.

The second contribution builds on the first. I hope to establish that the dynamics that structure households and families under capitalism are global dynamics. Just as the productive dynamics of the US auto industry were dependent on a supply of cheap

rubber, so too the households in which its high-waged factory labor was reproduced were dependent on social reproductive dynamics in the periphery that deflated the value of the rubber workers' labor. In a very real sense, the family-based housing policy in early 20th century industrial communities, as pushed by both firms and the state, contains a sort of negative space, a policy of household destruction both in racialized communities at home and among working populations further down the commodity chain. The working-class consuming nuclear family of the early twentieth century was only possible - could only be paid for - if raw materials from abroad were sufficiently cheap to make consumer goods such as cars profitable at a low cost. This cheapness stems largely from the low cost of labor, which was maintained in part by a systematic dissolution and reconfiguration of household structures at the point of production for foreign imports.

In examining the articulation between households in automotive factories and those in the rubber fields, I will be paying special attention to housing policy. Many studies of housing dynamics take the perspective of commodification: i.e. they ask what effect a home's dual nature as both a place to live and a financial instrument has on its residents. I would like to take a complementary approach and examine the home's role in disciplining working families and organizing their labor. For the part of our investigation focusing on automotive factories, this means we will look at the social programs of the Ford Motor Company and General Motors, such as Ford's Five-Dollar Day, and GM's Modern Housing Corporation. These policies helped create a home-owning class of well-paid workers who were tied to their local factories by mortgages.

If we take seriously the suggestion that this was possible (in part) because of a depressed value of labor in rubber fields and thus depressed rubber prices, then we are forced to ask what kinds of housing and social policies maintained this price suppression in Brazil and Malaya. To answer this we will look at the state negotiations of living and working conditions on Malayan rubber plantations, and the living conditions and reproductive dynamics of rubber harvesters in the Amazon. The next section will briefly outline some theoretical approaches to the household and their global context that will aid my study.

Background

The Household

The world-systems approach to the household stems from the insight that proletarianization, a fundamental process of historical capitalism, does not operate at the level of the individual worker. Proletarianization can be broadly understood as the historical constitution of a mass of 'free' labor - free in the sense that workers are not compelled to work by law or force, but are free to choose between work and starvation. The worker and employer are legally equal in this scenario, and the compulsion to work is mediated through the market and the labor contract, rather than through the whip. This shift from direct to indirect domination (or personal to impersonal, or political to economic), is accompanied by an untethering of the worker from the land. No longer a peasant, the proletarian worker reproduces him- or herself not directly, through subsistence labor in familial lots, but indirectly, through the exchange of wages for the fruit of another's labor in the general commodity market.

Immanuel Wallerstein notes (along with many many others) that “in historical capitalism, there has been an increasing proletarianization of the workforce” (Wallerstein 1983, 22), and yet the ideal, purely proletarian worker - if s/he has ever existed - is exceedingly rare. After four or five hundred years of capitalist development, “what is surprising is not that there has been so much proletarianization, but that there has been so little” (Wallerstein 1983, 23). Indeed, every hour of labor purchased on the market for a wage and exploited by a capitalist is supported and reproduced by a whole matrix of “unproductive” activities. This matrix includes cooking, cleaning, childcare and other often feminized forms of work, but it also includes schooling, healthcare, welfare, public infrastructure such as parks and roads and even the family garden plots that have been historically common in working class households. The capitalist laborer may have to rely on the market for much of the stuff of life, but this reliance is never complete, and is always supported by some piece of uncommodified existence.

For world-systems analysts, the articulation between waged labor and the non-waged activities that support and reproduce it are best understood as taking place within a household: a “framework of relatively stable structures which share a command fund of current income and accumulated capital” (Wallerstein 1983, 23). Income here is understood broadly, and refers to waged and rental income as well as subsistence activity, petty commodity production, and cash transfers from charity organizations or the state. Wallerstein notes several challenges to conceptualizing and demarcating households: “the boundaries of these households are constantly changing by the entries and exits of individuals”; the “household has usually been a kin-related unit, but

sometimes not or, at least, not exclusively”; and it has “for the most part been co-residential, but less so as commodification proceeded” (Wallerstein 1983, 24).

These challenges must not be underestimated or avoided, but embraced: it is precisely these challenges that motivate this investigation. My project will examine the dynamics of the household in situations that only marginally resemble the western nuclear family. In automotive Michigan, the nuclear family reigned supreme, but it was supported by a vast network of gendered and racialized reproductive labor - often coming not from household members, but from members of the substrata of domestic and service workers. The social relations that make up households become even more complex when we look at workers in the Global South. In the Amazon, work in the rubber fields was limited by season, and so laborers fell into patterns of transhumance in which household and reproductive dynamics were continually dissolved and reconfigured to fit the collective needs of harvesters hundreds of miles from home. And in colonial Malaya, the conditions of labor and reproduction were similarly attenuated by distance, as rubber plantations largely sourced labor from south Indian migrant workers. The Malayan case had the added complication of state interference, as negotiations between the governments of India and Malaya established the working and living conditions of these migrant workers.

The household is a central unit of analysis in world-systems analysis, but it is also studied widely by scholars outside world-systems tradition. In the last decade, the family and household have been the subject of significant theoretical and empirical work by social reproduction theorists such as Tithi Bhattacharya, Susan Ferguson, and Nancy Fraser (Bhattacharya 2017). Social Reproduction Theorists build on the work of

Lise Vogel in *Marxism and the Oppression of Women* (2013), linking the subjugation of women to their role in the generational reproduction of labor. One of Vogel's key insights is to place this role in a larger social dynamic of the reproduction of labor power that includes prisons, barracks, and even schools and hospitals. The situation of the Amazon and Malaya further demonstrate that a concept of the reproduction of the conditions of labor should be expanded to include forced and voluntary migration, slaving raids, and dormitory style plantation housing. My work aims to contribute to this literature by analyzing the role and limits of the household in securing labor pools, and by examining the articulation between families and households across regional and national borders.

The Modern World-System: The Core, The Semi-Periphery, and the Periphery

The dynamics that I wish to investigate are historical. As I examine the household and the social reproduction of labor, I must place them within a specific historical context: global capitalism. In doing so, I follow the example of dependency theorists and world-systems analysts who insist on placing local phenomena within the context of a global capitalist system. Wallerstein and Arrighi, two of the most prominent exponents of world-systems analysis, identify the beginning of the twentieth century as a crucial turning point in historical capitalism, when patterns of global exchange shifted and the US began to assume its role as global hegemon - a role it would occupy for most of the twentieth century. During this turbulent time in world finance, local moments of contestation over production and reproduction responded to and in turn influenced global dynamics. This project will situate the household's role in social reproduction

within the early automotive industry by examining the early automotive commodity chain, reaching from factories in Michigan to rubber fields in the Amazon and in Colonial Malaya.

Fordist historiography often treats Fordism as a dynamic that was exported to the peripheral economies in the mid 20th century, after being developed in America and Europe during and just after the World Wars. This is partly true, if we define Fordism to be a certain rationalization of production, but I contend that Fordist production practices had an immediate and significant interaction with the Global South from the very beginning of the automotive industry. Drawing on world-systems literature on commodity chains, my research takes as axiomatic that production processes cannot be analyzed within the boundaries of a single nation, but must be placed within a global context of production that utilized various configurations of labor. If this is the case, what were the forms of labor in the Global South that contributed to the production of automobiles, and how were these types of labor constituted and reproduced?

In *The Modern World-System I*, Wallerstein makes the case that income inequality and the uneven distribution of surplus is a defining dynamic of global capitalism. This dynamic persists both locally and globally. In a system based on the accumulation of capital and the investment of surplus (i.e. a capitalist system), Wallerstein asserts that “the optimal situation for a local investing class would be to have access to profits from low wages in the periphery and further profits from medium (as opposed to high) wages in their own area” (Wallerstein 2011, 84). In other words, in order to capture more surplus value and to free it up for re-investment, capitalists must reduce local wages - but not so far as to damage the local market in which value and

profits are realized. This contradiction is alleviated by the more extreme exploitation of workers in the periphery.

Already in the 16th century, Brazil and other regions in the Americas such as Chile and Peru were included as peripheral zones within an interstate capitalist system centered on accumulation in Europe (Wallerstein 2011, 68). From the very start, they were central to European capitalism. Initially, Brazil produced sugar for the world economy, on the basis of slave labor imported from western Africa (Wallerstein 2011, 88-9). The direct and extreme exploitation of slave labor provided cheap calories for the working classes in the core. In this sense, South American production formed part of the infrastructure which produced commodified labor in the core.

In the late 19th century, as the global demand for rubber exploded, the Amazon (and to an extent the Congo) served as the global provider. In the Amazon, Peru and Brazil contributed the most production. Rubber had been in use in the Amazon for thousands of years, primarily molded into bottles or balls, but with the invention of vulcanization it developed valuable industrial uses. In the case of our study, it was used to make automotive tires. In the first decades of industrial rubber production, it was collected from wild rubber plants rather than being grown on plantations. In fact Henry Ford's attempt to start rubber plantations in Brazil ended disastrously, as monocrop rubber cultivation increased the spread of the South American rubber blight. In the Amazon, the harvesting season was short, and workers fell into a pattern of transhumance: living with their families in cities in the off season and then entering the collection fields during the working season. Workers primarily came from an agricultural

background, forced to look for other work by the drought of the late 1870s (Cardoso de Mello and Van Melkebeke 2019, 160).

In the 1910s, the global market for rubber collapsed, and wild-rubber collection declined in profitability - damaged in the Amazon by the rubber blight. In Colonial Malaya, however, rubber cultivation was on the rise, initially planted in small quantities by small-scale coffee farmers looking to shore up lost profits from a coffee price slump. Over the next decades, as the Malayan land market was opened up to foreign investment, small farms were consolidated into large ones, and then into plantations, and capitalists were faced with the issue of labor supply. This was resolved primarily by recruiting south Indian migrant workers who traveled to Malaya for a few years, sending money back home before returning to their families. Interestingly, the working and living conditions of these workers were negotiated between the colonial states of Malaya and India. As in Brazil and Peru, the families of workers were attenuated across distance, with the wage earning workers isolated from those they supported.

Amazonian and Malayan production, and indeed all capitalist production, connect to what are called commodity chains. Commodity chains are the linked production processes that produce commodities. Automotive commodity chains, for example, include the assembly of the finished car, but also the production of raw materials, their transport, mediate processing of the materials into automotive components, and crucially the reproduction of labor power at each of these points. Automotive companies were forced to source rubber from the Global South, and the long chains of transit added considerable cost to the final produced commodity. This is offset - indeed the

whole system is made possible - by the considerably lower price of labor in the Global South.

The development of global infrastructure in the capitalist world-system is a crucial element in the extension of commodity chains. I primarily conceive of infrastructure as the means by which the capitalist system maintains the global division of labor, and conditions the corresponding value differential in labor and goods. The incorporation of peripheral zones into the capitalist world-economy required the development of an extensive system of shipping routes, transit lines, and roads. This system connected markets in the core with cheap labor supplies in the periphery, but they also maintained the cheapness of these labor supplies. As a part of the study of labor regimes along commodity chains, we will have to take into account the innovations in infrastructure that supported and made them possible.

Methods

My methodological approach is informed by several traditions, but is basically animated by Moishe Postone's approach to critical social theory (1993). In his reinterpretation of Marx's critical theory, he attempts to historicize the central category of capitalism: labor. Rather than treating labor as a transhistorical category that is acted upon and mistreated by capitalists, he critiques labor as constitutive of capitalism itself. In this project, I aim to undertake a parallel analysis of the household: I do not treat the household as some domestic sphere separate from capitalism that is acted upon by capitalists or the state, but as itself a constitutive category of the global capitalist system. With the spread of capitalism, families were of course destroyed and traditional

bonds dissolved, but they were then recomposed according to capitalist logic. What are the specifically capitalist forms of households? What are the role of the state and the firm in developing and maintaining them? I hope to investigate why capital has such an intense interest in cultivating diverse types of households, and what purpose they serve in propping up global structures of exploitation and accumulation. I believe that the early automotive industry provides an ideal case to examine these dynamics.

In this project, I primarily draw on the literature of historical sociologists and marxists feminists to historicize the category of the household. I am especially influenced by the work of world-systems analysts who study the household as the unit of the reproduction of labor, and the various forms it takes in different parts of the world. At the core of this project, I have identified two related problems to investigate, and I will attempt to demonstrate here how I will investigate them empirically. The problems are: 1) automotive production in the core caused immediate changes in patterns of production in the Global South that do not conform to core-centric conceptions of Fordist production; and 2) the patterns of household reproduction in the periphery do not conform to conceptions of household-based labor renewal that are organized around a nuclear family. In selecting these issues, I assert that the cases of rubber producers in the Amazon and Colonial Malaya demonstrate the futility of isolating local phenomena from their global context, and more specifically that the household dynamics of labor reproduction are in fact global dynamics. I want to show, not merely that the nuclear family which reproduces formally free proletarian labor is but one among many possible household organizations, but that it is in fact dependent on the

existence of other forms of family organization which reproduce unfree conditions of labor.

My approach to the first problem, that of the global effect of early Fordism, will mostly rely on secondary sources, and the work of economic and labor historians. My main reference points for the domestic historiography of Fordism are the works of Stephen Meyer III and Nelson Lichtenstein on American automotive labor history, *On the Line* (Lichtenstein and Meyer III 1988). For literature on Fordism as a stage of capitalist development, I use Aglietta's regulation account of American capitalism, *A Theory of Capitalist Regulation* (Aglietta 1979), Harvey's discussion of Fordism in *The Condition of Postmodernity* (Harvey 1989), and Alfred Chandler's account of the rise of the management class in American business, *The Visible Hand* (Chandler 1977). With these and related works, I hope to establish Fordism as a revolution not only in the field of production (i.e. the development of the assembly line and the decomposition of productive tasks), but also in the dynamic of social reproduction. Henry Ford was concerned not only with the shop floor, but also with the private lives of his employees, as demonstrated by his sociological association, his Five Dollar Day, and his interest in educating and 'Americanizing' immigrant laborers.

This view of Fordism is widely accepted, though it is not necessarily universal. Problems begin to arise when scholars investigate the relevance of Fordism to other parts of the world. Important, if incomplete, attempts include Alain Lipietz's essay "Towards Global Fordism" (1982), and Arrighi's discussion of the US-led cycle of accumulation in *The Long Twentieth Century* (2010). Lipietz critiques the idea that, in the 1980s, the industrialization of the Third World could be considered Fordist. Although

the search for cheaper wage-zones outside the capitalist core “develops an intrinsically Fordist mechanism [by] spreading ‘branch circuits’ over a pool of unevenly skilled and unevenly paid workers” (Lipietz 1982, 37), the social formation of the periphery and semi periphery deviates significantly from that in the Fordist core. Rather than relying on the economic and indirect compulsion of the wage form, in the Third World, capital must discipline labor through regulation, repression, and regimentation (Lipietz 1982, 43). Lipietz argues that a better name for this ‘exportation’ of Fordism is ‘Bloody Taylorism’.

But what of the immediate effect of Fordism on the rest of the world? Arrighi’s approach to Fordism is crucial to this study, in part because he insists on placing it within its immediate global context. Arrighi follows Braudel’s conception of capitalism as a three tiered structure: world trade rising above local markets and daily life. This informs his methodology, but also limits the scope of his project, as he says: ‘[o]ur analytical construct, therefore, focuses on that top layer and offers a limited view of what goes on in the middle layer of market economy and the bottom layer of material life’ (Arrighi 2010, 25). In this analysis, Fordism is a moment in the logic of the rise of US hegemony on the global field. Arrighi notes that “by 1902 Europeans were already speaking of an ‘American invasion’; and by 1914 US direct investment abroad amounted to 7 per cent of US GNP” (Arrighi 2010, 249), as lean American companies powered by a scheme of flexible production, increased consumption, and welfare policies, combined with the vertical integration of production inputs began to spread overseas. My project aims to take Arrighi’s analysis as a starting point and head back down to the level of daily life, and to investigate the interaction between these global dynamics and the reproduction of life in the rubber fields of the Amazon and Malaya.

This brings us to the second question, that of the types of 'household' which serve as the basic unit of the reproduction of labor. This category is closely linked to that of the 'family', which is often invoked as occupying the same structural position as household: the source of the regeneration of labor power. Important works discussing this dynamic include Wally Secombe's *Weathering the Storm* (1995), and the volume edited by Joan Smith and Immanuel Wallerstein, *Creating and Transforming Households* (1992). All of these works examine the structural role of households and families within capitalism. By providing a steady supply of labor, families and households enable capitalist accumulation. Wallerstein and Smith additionally focus on the role of the state in disciplining capitalist households.

Lise Vogel, however, sees the reproduction of labor as a dynamic that far exceeds the family - or rather sees the family as just one of the many institutions used by capital in order to secure and discipline sources of labor. Other institutions include prisons, barracks and work camps, but also hospitals and schools. Vogel's expanded scope provides us with a more flexible perspective through which to examine how labor power is secured and disciplined by industry. My project aims to use these two insights to investigate the reproduction of rubber-labor in the Amazon and Colonial Malaya. What was the role of the state in disciplining the reproduction of labor pools, and to what extent was this an extra-familial process? This investigation will be based mainly on primary sources, including Brazilian, Peruvian, and Malayan labor legislation, population and migration records, and any recorded testimonies of rubber workers.

In the case of Tamil workers in Malaya, the colonial state created vast amounts of documentation about the migration process and the working conditions in Malaya

(Amrith 2010). While these documents will be valuable, and will form the backbone of this portion of my project, I will follow Chakrabarty (1989) in examining these documents critically, looking for whatever 'optical errors' were reproduced in them as a result of the biases inherent in the state's viewpoint. I will bring a similar perspective to state documents pertaining to the rubber fields of the Amazon.

Sources of Evidence

This project is organized around three moments in the reproduction of labor along the automotive commodity chain: rubber collection in the Amazon, rubber cultivation in Colonial Malaya, and the final assembly and manufacture of automobiles in Michigan, USA. Each of these moments will be investigated through archival research, focused on emergence of different forms of working class household: the establishment of a nuclear family in Michigan, the analysis of transhumant workers in the Amazon, and the analysis of the role of the state in establishing working conditions for migrant laborers in Colonial Malaya.

There are extensive archives in Michigan that document the working and living conditions of workers in the automotive factory. Foremost among these are the General Motors Heritage Archive in which I hope to find internal documentation regarding the administration of the Modern Housing Corporation, and the Sloan Museum Archive which contains materials related to community life among General Motors workers. I also hope to visit the Benson Ford Research Center and gain access to their archives on the Ford Motor Company's social projects including internal reports for such programs as their english schools, and their profit sharing initiatives.

The Benson Ford Research Center also contains extensive documentation of Ford's failed rubber plantation project in Brazil: Fordlandia. This project was reported on widely in the local press, and so the newspaper archives in the Detroit Public Library are valuable, but mostly I am interested in the papers of the Benson Ford archives that establish logistical and material links between Henry Ford's social experiment in Brazil and its domestic production goals.

Other archival sources for rubber labor conditions in the Amazon include the Goodyear Tire and Rubber Company records housed at the University of Akron. Additionally, I will examine the report of the Casement Commission on working conditions in the Peruvian Amazon, documents related to the Peruvian Amazon Company, as well as CE Akers' *Report on the Amazon Valley* (1912). Some of these documents are made available online, but others, such as the papers of British rubber companies, are held by the Bodleian Library in the UK. I will emphasize identifying the economic actors responsible for collecting and exporting rubber, and I will search for links to recorded first-hand and second-hand accounts of the lives of rubber workers.

The portion of the project that deals with migrant laborers in south Asia will be based on documents produced by the Colonial state in India and Malaya. For the first half of the twentieth century, roughly corresponding to our period, the British Crown published a series of what were called Colonial Reports. These were published in English, and contained economic and population statistics, as well as official proceedings concerning the governing of the colonial states. There are several series of annual Colonial Reports that will form the basis of our empirical investigation into Malayan labor conditions. These include the *Annual Report of the Agent of the*

Government of India in British Malaya, the *Annual Report on the Social and Economic Progress of the People of the Straits Settlements*, and reports on immigration statistics. Many of these reports have been digitized and made available online by various university libraries. An extensive collection of these reports is held in microfiche format by Monash University in Melbourne Australia. Another valuable source is the *Straits Settlements Government Gazette*, made available online by the Singaporean government.

Research Plan

This is a large project, as it is no small task to research three distinct geographical regions. In this section I hope to lay out a reasonable research plan that will allow me to complete this project by the end of 2026. Despite its size, there are three things that I believe will help me to finish this project on time: 1) I have already performed research related to this project while here in the Binghamton Sociology Program; 2) the archives I need to visit have been increasingly digitized and made available online, thus reducing the need for expensive and time consuming travel; and 3) the Sociology Department at Binghamton has excellent mechanisms of support, some of which I have already received such as the Nesar Ahmad Memorial Scholarship, and some of which I hope to apply for in the future, such as the Dissertation Writing Fellowship.

For the academic year 2024-2025, I applied for and received the Nesar Ahmad Memorial Scholarship, housed in the Binghamton Sociology Department. Using this funding, I plan to spend this year researching the living and working conditions of Tamil

migrants in Malayan rubber plantations. As much of this documentation has been digitized and made available online, I do not plan to travel to south Asia. I will, however, use the funds I received to travel to Washington D.C. in the spring of 2025, to visit the Southeast Asian Reading Room in the Library of Congress, which contains English-language documents and periodicals from 20th century Malaya. I will also visit the South Asian Reading Room which contains archives of English language periodicals in Madras, such as *The Hindu*, and the *Madras Times*. Between now and then, I will search the Library of Congress catalog online to identify specific materials within the South Asian and Southeast Asian collections that I want to examine.

I have already been working on the domestic portion of this project for several years. When I first joined the Binghamton Sociology Department, my project was focused exclusively on the housing policies of the Ford Motor Company and General Motors. Although the scope has expanded significantly, research that I undertook during my first two years in this program is still highly relevant. In the winter of 2021, during the Covid lockdowns, I corresponded with archivists for General Motors, and for the city of Flint, Michigan, and received access to limited documentation regarding General Motors Modern Housing Corporation. In the summer of 2025, I hope to travel again to Detroit and Flint to supplement this research with in-person visits to the archives listed above, and to review documents related to Henry Ford's Fordlandia project.

The last part of the project that I will undertake is the study of households among rubber workers in the Amazon. This is partly to give me time to continue learning Spanish, which will be helpful in examining primary sources in Peru. At present, I am studying Spanish independently, but I plan to enroll in UW Madison's course in Spanish

Reading and Translation in the Summer of 2025. It is an online course, and I will seek out funding from Binghamton's Graduate Student Organization to cover the tuition.

However, the main sources I will be using for the portion of my project that is centered on Amazonian rubber production will be in English. Many of these are available online, such as Roger Casement's report on slavery in Peruvian rubber companies. Others will require me to travel: the Bodleian Library at the University of Oxford contains extensive papers from the British Anti-Slavery society, which researched and wrote about the Putumayo genocide perpetrated by the Peruvian Amazon Company, as well as papers from the Tambopata Rubber Syndicate.

Traveling to the Bodleian library will be the most significant research trip I undertake on this project. In order to make this research possible, I plan to apply for a Dissertation Writing Fellowship for the spring semester of 2026. In conjunction with other funding opportunities I plan to apply for, this will allow me time to visit the necessary archives, and write up my findings. If all goes well, then I hope to spend the summer and fall of 2026 finalizing my dissertation.

Proposed Chapters

1. Introduction: Fordism in Global Perspective

This chapter serves two main purposes: to introduce the questions I will be investigating, and to explain the theoretical framework I will employ to answer them. I will begin by surveying the existing literature around Fordism and its implementation in the Global South starting in the 1960s. Using the concept of commodity chains, I hope

to complicate this narrative and demonstrate that the development of Fordist practices in the early automotive industry required an immediate restructuring of productive relations in the Global South, especially in the rubber producing regions. From here I will introduce my main research question: if Fordism developed a new norm of working class nuclear households in Michigan, what effect did it have on households in other parts of the world? Or more pointedly, in what ways did the reconfiguration of peripheral households make the American nuclear family possible?

2. Modern Housing Corporation

This chapter will investigate the domestic dynamics of social reproduction in the Michigan automotive industry from roughly 1914 through the mid 1930s. I will first focus on the Five Dollar Day program implemented at the Ford Motor Company, with attention to housing policy. I hope to establish the firm's interest in propagating single income households among their white workers, and to question why this was never generalized to the entire population. I will then look at the Modern Housing Corporation, a division of General Motors which built and sold homes to its employees. This program differed from earlier company town models in that it fostered home ownership among workers rather than reserving all property rights with the firm. As with the Five Dollar Day, this program was administered on a segregated and racialized basis, and contributed to the development of a deeply stratified workforce in which the value of Black labor was depressed.

3. Fordlandia

This chapter will look at Henry Ford's attempt to start a rubber plantation in the Amazon in the 1920s. His plantation, which was a colossal failure, was based on the notion that the social-reproductive regime in Michigan could be imported to other parts of the world. Fordlandia was a mirror image of working class neighborhoods in the US, complete with prefabricated houses, schools, parks, and libraries. The administered life in Fordlandia met with resistance from local workers in Brazil and the town was abandoned in the mid 1930s without producing any commercially significant amount of rubber. This chapter will attempt to answer the question: why was the establishment of a town based on Fordist productive and reproductive norms unsuccessful in Brazil? In what ways did Ford misunderstand his own innovation, and why were his production and social practices unfit for rubber production in Brazil?

4. Transhumant workers in the Amazon

Although Ford's rubber plantation failed to be productive, the Amazon as a whole led the world in rubber production in the late 19th century and early 20th. This chapter will examine the patterns of wild-rubber collection in the Amazon and the household formations that supported them. First I will look at transhumant rubber workers in Brazil who moved seasonally between the port cities and the rubber fields deeper inland. What are the social-reproductive dynamics of transhumant labor? As working men were separated from their families for a season, how did they arrange the daily renewal of their labor power, and how did their families deal with the long gaps in income? The chapter will then look at the Peruvian Amazon Company, a rubber firm based in Iquitos

that notoriously kept workers in near-slavery, and that carried out the Putumayo genocide. How was a continual supply of labor secured in the midst of mass slaughter?

5. Migrant workers in Colonial Malaya

In the 1910s, wild harvested Amazonian rubber was superseded in the global market by rubber produced on plantations in Colonial Malaya. This chapter will chart the rise of Malayan rubber, as well as the development of its specific labor regime. The workers on Malayan rubber plantations were primarily ethnic Tamils from Madras who migrated to Malaya for multi-year labor stints. I hope to trace out their working and living conditions on the plantations, and to contextualize this within the broader context of their household groups. As with transhumant workers in Brazil, I will ask how daily life was reproduced among the largely male workers at the work sites, and what effect this migratory pattern had on their families who were left behind. Did communities in Madras have to adjust their structure to make up for lost laborers? How were funds transmitted between workers and their families, and what were these funds used on? In other words, how did households handle being stretched across the Bay of Bengal?

6. Conclusion: Beyond Labor, or the Social Lives of Workers

This chapter will examine the limits of my project, and indicate potential ways to overcome them. By necessity this chapter cannot be fully sketched out at this point: the failures and shortcomings of the project will emerge through research and writing. I can, however, point to one important consideration that this project does not cover.

Wallerstein writes that in addition to being an income-pooling unit, the household is also the site of ideological conditioning, in which children become conditioned to their place within the world-system. Similarly, Kirstin Munro writes that the household is not merely the site of the social reproduction of labor, but a fundamental institution in the reproduction of capitalism itself (Munro 2019). In this conclusion, I will examine the ways in which my research addresses these additional functions of the family, and I will indicate potential further avenues to research these dynamics within the early automotive industry.

Annotated Bibliography of Secondary Sources

Automotive Industry in the Imperial Core

Aglietta, Michel. 1979. *A Theory of Capitalist Regulation: The US Experience*. London: NLB.

This history of American capitalism was influential among French historians of the Regulation school. Aglietta's book argues that American capitalism underwent a profound change in the 20th century, typified by the Fordist emphasis on both production and consumption. The focus on consumptive habits of workers is crucial, but Aglietta's perspective is limited by its methodological nationalism, and his dismissal of the crucial role of imperialism in American industry.

Arrighi, Giovanni. 2010. *The Long Twentieth Century: Money, Power, and the Origins of Our Times*. 2nd ed. London: Verso.

In the final chapter of this book, Arrighi discusses the rise of US hegemony. He argues that this is partly based on several innovations in US industry: the development of a management class, vertical integration, and the internalization of transaction costs. This argument places the US automotive industry in a world-historical context, and isolates the key characteristics which extend beyond US borders.

Harvey, David. 1989. *The Condition of Postmodernity*. Oxford, England: Blackwell.

Chapter 8 of this work, titled simply “Fordism” is the most relevant to my project. Harvey discusses the relationship between Fordism and Taylorism, and Gramsci’s analysis of them both. He treats Fordism typically: as a set of productive relations that was developed in America and then exported to Europe after the second world war. Harvey is clear on the stratified and exclusionary nature of Fordist development in the US and Europe, but discussion of Fordism in the Third World is relegated to a single vague paragraph at the end of the chapter.

Lichtenstein, Nelson, and Stephen Meyer III, eds. 1988. *On the Line: Essays in the History of Auto Work*. Urbana: University of Illinois Press.

This book is a series of critical essays on the history of automotive production. Although limited in scope to America and Europe, essays in this book make two crucial contributions toward understanding Fordism. First, they challenge the

notion that Ford was successful because industrial Michigan lacked a unionized workforce, and emphasize the extent to which Ford and other firms manipulated the existing labor force to their own needs. Second, they challenge the narrative, typified by Braverman, that Fordism resulted in a uniform deskilling of the workforce. Rather they note that for a small portion of labor, the level of skill required increased dramatically. This concept of a stratified labor market is crucial for my understanding of the global automotive industry.

Meyer III, Stephen. 1981. *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921*. Albany: State University of New York Press.

This book is a classic account of Henry Ford's social policies. Meyer argues that a key innovation of early Fordism, the Five Dollar Day, was part of a program of social control. It demonstrates the extent of capital's interest in the daily lives of its employees, and enumerates the specifics of this interest in the early automotive industry.

Rubber Industry in the Periphery

Bunker, Stephen G., and Paul S. Ciccantell. 2005. *Globalization and the Race for Resources*. Baltimore: Johns Hopkins University Press.

This book investigates the development of extractive industries in the Amazon. Paying special attention to the mining of iron, Bunker and Ciccantell trace the growth of an infrastructure that integrates remote sites of raw materials with the global market. This book will be useful both generally, in its explanation of the subjugation of local processes to global economic needs, and in the specifics of extraction in the Amazon. The second chapter of the book deals with the development of the rubber industry in the Amazon, and the technological developments that made it possible.

Coates, Austin. 1988. *The Commerce in Rubber: The First 250 Years*. Oxford: Oxford University Press.

A valuable account of the global rubber trade. This book surveys the history of the rubber industry from wild rubber trees in the Amazon and the Congo to plantations in south Asia, to synthetic production in the mid to late twentieth century. It identifies major firms involved in the production and sale of rubber and the global markets they operated within. Coates also explains the major technologies such as vulcanization that developed to make rubber production possible on a large scale. The book does not deal comprehensively with labor conditions in the rubber industry, nor does it deal with the effects of rubber cultivation on local populations in any great detail.

Frank, Andre Gunder. 1967. *Capitalism and Underdevelopment in Latin America: Historical Studies of Chile and Brazil*. New York: Monthly Review Press.

Frank's work on Latin America was crucial in the development of both dependency theory and world-systems analysis. His concept of the development of underdevelopment challenged the developmentalist paradigm and insisted that the underdevelopment of peripheral countries was actively cultivated by core countries. This book examines Brazil and Chile in this context, tracing the effect of global economic polarization and extraction on those two countries.

Headrick, Daniel R. 1988. *The Tentacles of Progress: Technology Transfer in the Age of Imperialism, 1850-1940*. Oxford: Oxford University Press.

Tentacles of Progress attempts to answer, in part, why the poorer countries had not become industrialized by the mid twentieth century. The answer lies in the relationship between the richer and poorer countries, which held the poorer economies in a state of underdevelopment. This was effected with help from specific technologies: railways, shipping routes, telecommunications, etc. Headrick examines both the social context of deployed technologies and the technological context of emergent social formations, with attention paid to the 'tropical economies' such as south Asia and Africa. It is in this socio-technological context that our study takes place.

Parmer, J. Norman. 1960. *Colonial Labor Policy and Administration: A History of Labor in the Rubber Plantation Industry in Malaya*. Monographs of the Association for Asian Studies, IX. Locust Valley, NY: J. J. Augustin Incorporated Publisher.

Colonial Labor Policy surveys the labor conditions of Malayan rubber plantations from 1910 - 1941. Based on extensive archival research, this book deals with the rise of the rubber industry in Colonial Malaya. It deals with both native and immigrant labor, and with the relation between the state and capital in developing the industry. This book will help me identify primary sources, and will help to contextualize them within Malayan industry. However, the book is limited in that it is drawn mainly from official documents. It needs to be supplemented with social archives, and with a more robust contextualization of Malaya within the global economy.

Households and Families under Capitalism

Secombe, Wally. 1995. *Weathering the Storm: Working-Class Families from the Industrial Revolution to the Fertility Decline*. London: Verso Books.

A comprehensive account of changing family dynamics under capitalism. Secombe combines demographic analysis with a marxist feminist perspective to position the family as a constitutive social institution of capitalism. The book poses a powerful challenge to the notion that the nuclear family is a transhistorical social unit, but it is severely limited by its focus on the imperial core. I hope to extend Secombe's analysis to the periphery, and to test his conclusions against the situation of the Global South.

Smith, Joan, and Immanuel Wallerstein, eds. 1992. *Creating and Transforming Households: The Constraints of the World-Economy*. Studies in Modern Capitalism. Cambridge: Cambridge University Press.

This book is an edited volume of essays analyzing the shifting structures of households in the world-economy. It is valuable for the theorization of households as income-pooling units, and for the chapter on Fordism and Detroit. This chapter challenges the notion of the family wage as the generalized state of American households, carefully enumerating the ways in which households rely on varying amounts of subsistence labor depending on their position within the stratified local economy.

Vogel, Lise. 2013. *Marxism and the Oppression of Women: Toward a Unitary Theory*. Revised Edition. Historical Materialism Book Series 45. Leiden, Netherlands: BRILL.

Vogel's book traces arguments on the left as to the source of women's oppression. She rejects the so-called 'dual system theory' that separates patriarchy from capitalism, insisting instead that at present women's oppression is inseparable from class oppression. She situates the oppression of women within a broader social phenomena: the reproduction of labor. Her linking of generational reproduction to the daily reproduction of labor in work camps, prisons, and barracks is crucial to my study. More attention will need to be paid to the roles of immigration and slavery in securing and reproducing the conditions of labor.

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Amrith, Sunil S. 2010. "Indians Overseas? Governing Tamil Migration to Malaya 1870-1941." *Past & Present* 208 (1): 231–61.

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Cardoso de Mello, Louise, and Sven Van Melkebeke. 2019. "From the Amazon to the Congo Valley: A Comparative Study on the Violent Commodification of Labour During the Rubber Boom (1870s–1910s)." In *Commodity Frontiers and Global Capitalist Expansion*, edited by Sabrina Joseph, 137–81. Palgrave Studies in Economic History. Palgrave MacMillan.

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